

Thermodynamic Analysis

Worst, Normal, and Best-Case Thermal Throughput, Power, and Solar Coverage

Nebico Oil Storage — Adaptive Demand Allocation Architecture (ADA)

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This is an analysis paper, not a design proposal under review. It evaluates thermal throughput, coil power consumption, solar coverage, and estimated cost savings under three environmental/ feedstock scenarios — Worst, Normal, and Best case — using the water-side hydraulic basis established in Architectural Revision II (2-inch coil bore, 1.2 m s^{-1} feed velocity, $70\text{ }^{\circ}\text{C}$ constant water supply) and the full three-way concurrent branch architecture.

Executive Summary

Metric	Worst	Normal	Best
T ₅ duty period (hr)	4.70	4.03	3.64
PO tank duty period (hr)	0.91	0.45	0.00
Palm oil melted (kg/day)	5457.78	5439.29	5490.83
Total daily energy (MJ)	1247.52	1173.58	1133.62
Average coil power (kW)	14.44	13.58	13.12
Solar (FPC) daily yield (kWh)	22.35	29.82	29.78
Solar coverage of demand	6.45%	9.15%	9.46%
Estimated daily savings (NPR)	205.8	274.6	274.2
Estimated annual savings (NPR)	75,125	100,243	100,094

- **Best case genuinely requires zero PO tank duty** – starting at 35 °C, olein is already above its 30 °C target before any heat is applied. This is a real finding of the analysis, not a simplification.
- **Solar coverage is materially lower than earlier revisions reported** (6.4–9.5% here vs. 40–61% in the earlier ADA Architecture proposal) – driven by a smaller FPC array (10 panels, 19.0 m², vs. the original 15-panel, 22.674 m² array) against a substantially larger total daily energy demand under the full three-way-concurrent throughput basis.
- **Worst case is the only scenario materially different in collector efficiency**, dropping to 34.0% (vs. $\approx$ 45.3–45.4% for Normal/Best) purely from the wider ΔT against 2 °C ambient.

1 Scenario Definitions

Table 1: Worst/Normal/Best condition parameters.

Parameter	Worst	Normal	Best
Ambient temperature, T_a	2 °C	18.1 °C (Kathmandu weighted avg.)	18 °C
T ₁₀ oil starting temperature	2 °C	18.1 °C	35 °C
PO tank oil starting temperature	2 °C	18.1 °C	35 °C

Best case represents a system that has been running continuously for a day – residual heat leaves feedstock significantly warmer than a cold start, while ambient remains at its typical Kathmandu value. Worst case represents both an extreme cold snap *and* a cold-start fill simultaneously; Normal case uses the Kathmandu annual weighted average for both ambient and feedstock, consistent with every prior ADA-family document.

The 35 °C Best-case feedstock starting temperature is an engineering assumption, not an exact calculation. It is an intuition-based estimate of residual warmth after a day of continuous system operation, not a value derived from a formal thermal-retention or heat-loss calculation. It should be treated as a reasonable planning figure rather than a rigorously computed one.

1.1 T₅'s Task (Fixed, Condition-Independent)

T₅ heats 4000 kg of oil from 35 °C to 55 °C in every scenario – this task itself does not vary by condition; only T₅'s standing loss (ambient-dependent) does.

1.2 Delivery and Target Temperatures

T₁₀ delivers oil to T₅ at exactly 36 °C – bare completion of melting, no further margin. The PO tank's target remains 30 °C, but its *starting* temperature now follows the same three-condition basis as T₁₀, rather than being held fixed at ≈ 24 °C as in earlier revisions.

2 Governing Basis (Retained from Architectural Revision II)

Table 2: Water-side capacity rates, unchanged from Architectural Revision II.

Duty	εC_w (W/K)
T ₅ (liquid)	205.3
PO tank (liquid)	570.5
T ₁₀ (solid, melt)	181.5

Coil bore fixed at 2-inch across all tanks, feed velocity 1.2 m s^{-1} , water supply constant at $T_{\text{hot}} = 70^\circ\text{C}$ – all as established in the attached Revised System Parameters excerpt. $C_{p,\text{solid}} = 2000 \text{ J kg}^{-1} \text{ K}^{-1}$, $C_{p,\text{liquid}} = 2100 \text{ J kg}^{-1} \text{ K}^{-1}$, $L_f = 150\,000 \text{ J kg}^{-1}$ (palm oil latent heat of fusion), matching every prior document in this project.

T₅'s standing-loss basis ($U = 0.686 \text{ W m}^{-2} \text{ K}^{-1}$, $A = 14.25 \text{ m}^2$) and the PO tank's lumped uninsulated loss coefficient ($UA \approx 59.08 \text{ W K}^{-1}$) are both reused directly from earlier established figures. The PO tank coefficient in particular was originally linearised around a narrow reference ΔT ($\approx 8.9 \text{ K}$); applying it across the wider ΔT range this analysis now spans (worst case oil averaging 16°C against 2°C ambient) carries the same approximation caveat flagged when it was first derived.

3 Specific Energy per Condition

Table 3: T₁₀ melt specific energy, $q = C_{p,s}(36 - T_{\text{start}}) + L_f$.

	Worst	Normal	Best
T_{start} ($^\circ\text{C}$)	2.0	18.1	35.0
q_{T10} (kJ/kg)	218.0	185.8	152.0

The Normal-case figure (185.8 kJ/kg) matches the earlier ADA Architecture document's original specific-energy figure exactly, confirming this formula is applied consistently with prior work.

Table 4: PO tank specific energy, $q = C_{p,l}(30 - T_{\text{start}})$.

	Worst	Normal	Best
T_{start} ($^\circ\text{C}$)	2.0	18.1	35.0
q_{PO} (kJ/kg)	58.8	25.0	0 (already above target)

4 Thermal Throughput Analysis

4.1 Derivation of the T_{10} Melting Window (FSM / HSM–LSM Split)

T_5 's and the PO tank's duty periods, t_{P1} and t_{P2} , are established first from their own sensible and loss energy requirements (tables below). Whenever the PO tank is active (t_{P2} hours), T_{10} is restricted to Full Supply Mode (FSM), delivering $\approx 98\%$ of nominal coil power to a single tank ($Q_{T10,FSM} = 6049$ W vs. 6176 W per tank in HSM/LSM). For the remainder of the day, $24 - t_{P2}$ hours, T_{10} runs in High/Low Speed Mode (HSM/LSM) across two tanks concurrently, so this remaining time is doubled to give the HSM/LSM equivalent-hour pool available for melting. T_5 's duty period is established independently and does not itself subtract from this window, since it runs on a dedicated line separate from the PO/ T_{10} branch split.

Table 5: T_5 (P1) duty period.

	Worst	Normal	Best
E_{sensible} (MJ)	168.0 (fixed task)		
E_{loss} (MJ)	44.76	32.4	25.34
E_{total} (MJ)	86.76	74.4	67.3
Q_{avg} (W)	5132.5 (fixed, midpoint 45°C)		
t_{P1} (hr)	4.70	4.03	3.64

Table 6: PO tank (P2) duty period.

	Worst	Normal	Best
E_{sensible} (MJ)	29.40	12.50	0
E_{loss} (MJ)	71.46	30.37	0
E_{total} (MJ)	100.86	42.87	0
Q_{avg} (W)	30,807	26,214	n/a
t_{P2} (hr)	0.91	0.45	0.00

Table 7: T_{10} (P3) remaining window and melt output.

	Worst	Normal	Best
Equivalent(hr) for HSM/LSM	46.18	47.1	48
Equivalent(hr) for FSM (1 Tank)	0.91	0.45	0
Q_{T10} (W, at 36 °C, FSM)		6049	
Q_{T10} (W, at 36 °C, HSM/LSM)		6176×2	
$E_{\text{available}}$ (MJ)	1059.90	1056.31	1066.32
Mass melted (kg/day)	5457.78	5439.29	5490.83

Colder starting feedstock has a limited net effect on daily throughput: worst case does carry the largest specific energy requirement per kg (218.0 kJ/kg vs. 152.0 kJ/kg for best case) and the smaller of the two melting windows (47.09 hr vs. 48.00 hr), since T_5 's and the PO tank's own losses are also worse in cold ambient. These two penalties largely offset one another in the mass-melted outcome, however: worst-case throughput (5457.78 kg/day) comes in at 99.4% of best-case throughput (5490.83 kg/day) – a difference of under 1%, not the compounding shortfall the specific-energy figures alone would suggest.

5 Power Consumption

Table 8: Total daily energy and average coil power draw.

	Worst	Normal	Best
E_{T5} (MJ)	86.76	74.4	67.3
E_{PO} (MJ)	100.86	42.87	0.00
E_{T10} (MJ)	1059.90	1056.31	1066.32
$E_{\text{day,total}}$ (MJ)	1247.52	1173.58	1133.62
P_{avg} (kW)	14.44	13.58	13.12

Average coil power is remarkably stable across all three conditions (13.1–14.4 kW) despite the underlying components shifting substantially – T_{10} 's melt energy dominates the daily total in every scenario (85–94% of the total), damping the relative swing that T_5 's and the PO tank's more condition-sensitive loss terms would otherwise cause.

6 Solar Power Coverage

6.1 FPC Configuration

$N = 10$ panels, each $2000 \text{ mm} \times 950 \text{ mm} = 1.9 \text{ m}^2$, giving $A_{c,\text{FPC}} = 19.0 \text{ m}^2$ – a smaller array than the original base-document design (15 panels, 22.674 m^2).

$$\eta_{\text{FPC}} = \eta_0 - \frac{a_1 \Delta T}{G} - \frac{a_2 \Delta T^2}{G}, \quad \eta_0 = 0.75, \quad a_1 = 3.50 \text{ W m}^{-2} \text{ K}^{-1}, \quad a_2 = 0.015 \text{ W m}^{-2} \text{ K}^{-2} \quad (1)$$

evaluated at $G = 750 \text{ W m}^{-2}$, $PSH = 4.61 \text{ hr}$, $\Delta T = T_{\text{hot}} - T_a = 70 - T_a$ (**note:** $T_{\text{hot}} = 70$ here, not the base document's 75, matching this document's fixed water supply).

Table 9: Solar (FPC) yield and coverage.

	Worst	Normal	Best
ΔT (K)	68.0	51.9	52.0
η_{FPC}	34.02%	45.39%	45.33%
Q_{peak} (W)	4847.7	6468.5	6458.9
E_{solar} (kWh/day)	22.35	29.82	29.78
E_{solar} (MJ/day)	80.45	107.35	107.19
Coverage of daily demand	6.45%	9.15%	9.46%

Coverage is substantially lower than the 40–61% figures reported in the original ADA Architecture proposal. Two compounding factors: (1) the FPC array here is $\approx 16\%$ smaller in area (19.0 vs. 22.674 m²); (2) total daily demand under the full three-way-concurrent throughput basis (1134–1248 MJ/day) is roughly 2–3 $\times$ larger than the demand figures earlier revisions were evaluated against. This is a direct, honest consequence of maximising throughput – it is not a solar-system sizing error.

7 Estimated Savings

Tariff sourced, not project-specific. Savings below use Nepal Electricity Authority’s general industrial/business rate, $\approx$ **NPR 9.21/kWh**, current as of the most recent published figures at the time of writing. Actual industrial tariffs in Nepal are commonly slab-structured and may include demand charges not captured by a single flat rate – this figure should be replaced with the facility’s actual negotiated/metered rate before this savings estimate is used for any financial decision.

Table 10: Estimated cost savings from solar (FPC) contribution.

	Worst	Normal	Best
E_{solar} (kWh/day)	22.35	29.82	29.78
Daily savings (NPR)	205.8	274.6	274.2
Annual savings (NPR)	75,125	100,243	100,094

Annual savings are strikingly similar between Normal and Best case ($\approx$ NPR 100,000 either way) despite Best case’s larger throughput – because solar yield tracks collector efficiency and ambient ΔT , not production volume. Worst case’s savings ($\approx$ NPR 75,125/year) are lower purely from the collector efficiency penalty at 2 °C ambient, not from reduced operating time.

8 Solar Power Scalability

Let x denote the total capital (NPR) allocated to a solar collector purchase, and let c_{FPC} , c_{ETC} denote the installed unit cost of each technology (NPR per m^2 of collector area – **still open, pending pricing**). The collector area purchasable for a given capital x is then

$$A_{\text{FPC}}(x) = \frac{x}{c_{\text{FPC}}}, \quad A_{\text{ETC}}(x) = \frac{x}{c_{\text{ETC}}}. \quad (2)$$

Both collector efficiencies are now fully characterised (Kathmandu conditions, $G = 750 \text{ W m}^{-2}$, $\Delta T = 70 - T_a$, consistent with the basis used throughout this document):

$$\eta_{\text{FPC}} = 0.75 - \frac{3.50 \Delta T}{G} - \frac{0.015 \Delta T^2}{G}, \quad \eta_{\text{ETC}} = 0.70 - \frac{1.50 \Delta T}{G} - \frac{0.005 \Delta T^2}{G}. \quad (3)$$

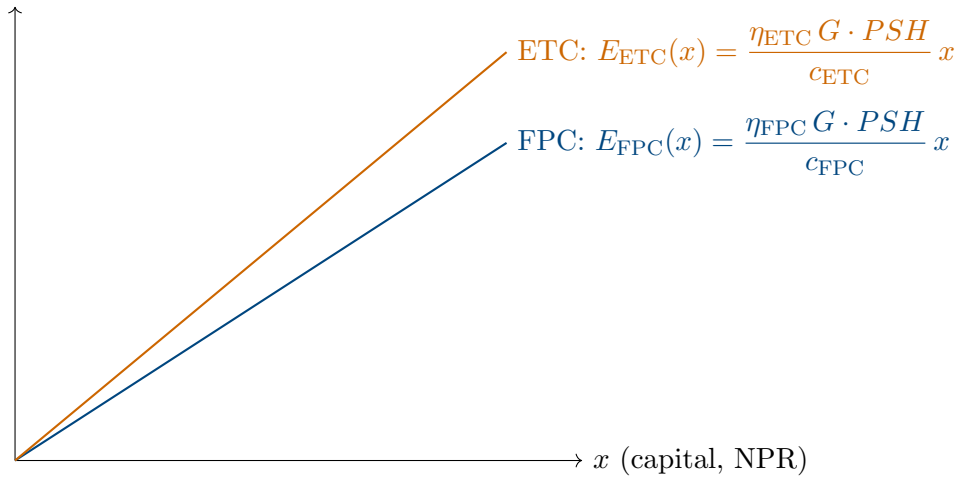
Table 11: ETC vs. FPC output on this document’s actual working area, $A = 19.0 \text{ m}^2$ (10-panel FPC array, established in Section 7).

	Worst	Normal	Best
ΔT (K)	68.0	51.9	52.0
η_{FPC}	34.02%	45.39%	45.33%
η_{ETC}	53.32%	57.82%	57.80%
E_{FPC} at 19.0 m^2 (kWh/day)	22.35	29.82	29.78
E_{ETC} at 19.0 m^2 (kWh/day)	35.03	37.99	37.97
Coverage of daily demand – FPC	6.45%	9.15%	9.46%
Coverage of daily demand – ETC	10.11%	11.65%	12.06%

Daily energy generation for a given capital allocation follows as

$$E_{\text{FPC}}(x) = A_{\text{FPC}}(x) \cdot \eta_{\text{FPC}} \cdot G \cdot PSH, \quad E_{\text{ETC}}(x) = A_{\text{ETC}}(x) \cdot \eta_{\text{ETC}} \cdot G \cdot PSH. \quad (4)$$

Energy generated (kWh/day)



Slopes shown are qualitative, not final. η_{ETC} is now a known, fixed value (53–58% across conditions, vs. 34–45% for FPC) – but the ETC line is drawn steeper here purely on efficiency; its true slope also depends on c_{ETC} , which is not yet priced. If c_{ETC} turns out sufficiently higher than c_{FPC} , the lines could cross or invert. The lines should be re-plotted from actual (x, E) data once both unit costs are available.

8.1 Unit Cost Analysis

Let x be the capital allocated to FPC and y the capital allocated to ETC. Defining ROI as the annualised value of generated energy at the tariff rate established in Section 7 (≈ 9.21 NPR/kWh) divided by capital outlay, the ratio of returns reduces to

$$\frac{\text{ROI}_{\text{FPC}}}{\text{ROI}_{\text{ETC}}} = \frac{\eta_{\text{FPC}}/c_{\text{FPC}}}{\eta_{\text{ETC}}/c_{\text{ETC}}} = \frac{\eta_{\text{FPC}}}{\eta_{\text{ETC}}} \cdot \frac{c_{\text{ETC}}}{c_{\text{FPC}}}. \quad (5)$$

The efficiency half of this ratio, $\eta_{\text{FPC}}/\eta_{\text{ETC}}$, is now known directly from the characterisation above:

Table 12: Efficiency ratio $\eta_{\text{FPC}}/\eta_{\text{ETC}}$ – the cost-independent half of the ROI comparison.

	Worst	Normal	Best
$\eta_{\text{FPC}}/\eta_{\text{ETC}}$	0.638	0.785	0.784

ETC is the more efficient collector at every condition in this project (0.638–0.785 $\times$ FPC’s efficiency, i.e. ETC generates 28–57% more energy per m^2 than FPC depending on condition). This means ETC only needs to win on cost by less than that margin – specifically $c_{\text{ETC}}/c_{\text{FPC}}\eta_{\text{ETC}}/\eta_{\text{FPC}} \approx 1.28\text{--}1.57$ – for it to also win on ROI. **This break-even threshold is the single number needed once both unit costs are priced;** no further recomputation of the efficiency side will be required.

9 Conclusions

1. **Best case's zero PO-tank duty is a genuine, load-bearing finding**, not a rounding artefact – olein starting at 35 °C is already past its 30 °C target before the coil engages at all.
2. **The full three-way-concurrent FSM/HSM–LSM operating window substantially outperforms every prior throughput figure in this project** (5439–5491 kg/day vs. 2447–5420 kg/day in earlier revisions), confirming the architectural value of true simultaneous branch operation established in the minor revision documents.
3. **Solar coverage is now a smaller fraction of total demand than earlier reported**, purely because total demand has grown faster than the solar array. This is worth flagging to stakeholders expecting the earlier 40–61% figures to still apply – they were computed against a much smaller throughput basis.
4. **The PO tank's loss-coefficient linearisation carries a wider approximation error in the worst-case scenario** than in any prior use of that constant, given the wider ΔT range now spanned – worth a full natural-convection recomputation if worst-case PO tank figures become load-bearing for a future decision.
5. **The savings estimate is directional, not final** – the tariff figure used is a general published rate, not this facility's actual metered/negotiated industrial rate.

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